Hyejin Yoon

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Bridging photonics and electronics through end-to-end research, from device design to full experimental system implementation

RESEARCH INTEREST

Photonic–Electronic Hybrid Systems, **Optoelectronics**, **Integrated Photonics**, **Fabrication-Driven Innovation**, **Quantum Photonics**, **Non-Hermitian Photonics**

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Republic of Korea
M.S. in Quantum Science and Technology (Transferred from Integrated M.S./Ph.D. Program) Mar 2024 – Feb 2027 (Expected)

Gwangju Institute of Science and Technology (GIST) Gwangju, Republic of Korea
B.S. in Electrical Engineering and Computer Science (Minor in Physics) Mar 2019 – Feb 2024

University of California, Berkeley California, United States
Fully funded exchange program Jun 2022 – Dec 2022

PUBLICATIONS

[J0] **H. Yoon**, ..., S. Kim **"InAs/GaAs Quantum Dots at 1D Exceptional Point Gratings"** (manuscript in preparation)

[J1] **H. Yoon***, S. Park*, Y. K. Kim*, J. Baek, K. H. Kim, S. Yun, H. Son, J. Choi, B. C. Jang, D.-H. Kang **"A Van der Waals Optoelectronic Synapse with Tunable Positive and Negative Post-Synaptic Current for Highly Accurate Spiking Neural Networks"** *Advanced Functional Materials*, 2026 (**Selected as Front Cover**)

[J2] S. Hasanli, M. Hasan, **H. Yoon**, S. Lee, S. Kim **"Exceptional Points in a Passive Strip Waveguide"** *Nanophotonics*, 2025

CONFERENCES

[C1] J. Song*, **H. Yoon***, S. Baek, S. Kim **"Second-Harmonic Microcomb Transfer in Dispersion-Engineered Concentric Ring Resonators"** (European Conference on Integrated Optics (ECIO), 2026)

[C2] **H. Yoon***, Y. Reum*, M. Lee, D. Shin, C. G. Mayer, J. S. Moon, M. Hartig, B. S. Damasceno, S. Höfling, S.-H. Gong, A. Pfenning, Y.-H. Ko, S. Kim **"InAs/GaAs Quantum Dots at 1D Exceptional Point Gratings"** (The 13th International Conference on Quantum Dots, 2026)

[C3] **H. Yoon**, J. Kim, S. Baek, S. Kim **"Phase-Mismatched Waveguide-to-Racetrack Microresonator Coupling"** (Optical Society of Korea Photonics Conference, 2024)

[C4] M. Park, **H. Yoon**, H. Park, Y. Park **"Self-Cavity SNSPDs for High-Efficiency Absorption"** (The Korean Vacuum Society annual conference, 2024)

[C5] **H. Yoon**, S. Park, D.-H. Kang **"MoS_2/InSe Heterostructure Optoelectronic Synaptic Device"** (The Korean Vacuum Society annual conference, 2023, **Best Poster Award**)

RESEARCH EXPERIENCE

KAIST Integrated Metaphotonics Lab, KAIST (Prof. Sangsik Kim) Daejeon, Republic of Korea
Graduate Researcher Mar 2024 – Present

- Led the full setup of a **PIC measurement platform** and designed experimental systems for **nonlinear optics and microcomb measurements**.
- Set up an asymmetric MZI-based measurement system and **performed real-time comb power measurement** and dispersion calibration using a photodetector and oscilloscope.
- Designed and optimized **ring–bus waveguide coupling** using analytical and numerical methods, performed resonator characterization for microcomb and frequency-conversion applications, developed heater-integrated GDS layouts, and submitted designs through a **Ligentec AN350 MPW fabrication run**.
- Analyzed and experimentally implemented **dispersion-engineered ring resonators**, demonstrated **frequency**

comb generation and **broadband SHG-based spectral transfer**.
- Designed and optimized **edge-coupled PIC measurement schemes** to improve **fiber-to-chip coupling efficiency**.

University of Würzburg Würzburg, Germany
Visiting Graduate Researcher (Host: Prof. Sven Höfling) Sep 2025 – Dec 2025
In collaboration with KAIST and Electronics and Telecommunications Research Institute (ETRI, Dr. Young-Ho Ko)
- Performed **COMSOL and RCWA simulations** to design and identify **grating structures supporting exceptional points**.
- Experimentally demonstrated **exceptional points** in InAs/GaAs quantum dot membrane photonic structures using **angle-resolved photoluminescence (ARPL)**, **time-resolved photoluminescence (TRPL)**, and **Fourier-plane imaging** under **cryogenic conditions**, revealing **LDOS enhancement and emission dynamics** in non-Hermitian photonic systems.

Nano Electronics and Optoelectronics Lab, GIST (Prof. Dong-Ho Kang) Gwangju, Republic of Korea
Undergraduate Researcher 2023 – 2024
- Designed and fabricated **2D-material-based optoelectronic synaptic devices** for neuromorphic applications.
- Performed **device fabrication**, **electrical characterization**, and **system-level analysis**.

SELECTED HONORS AND AWARDS

Outstanding Undergraduate Research Award GIST EECS, Dec 2023
2nd Place, Designing optoelectronic device for neuromorphic systems

GIST Innovative Convergence Technology Contest GIST, Aug 2023
1st Place, AI-based campus energy management system (EMS) leveraging high-performance AI models.

Academic Excellence Scholarship GIST, 2020 – 2022
Merit-based scholarship awarded for top **academic performance**.

Quantum Information Hackathon Ministry of Science and ICT, 2022
Community Award, recognizing for outstanding performance in quantum information applications.

Study Abroad Program Scholarship GIST, 2021
Fully funded exchange program at UC Berkeley. **(8 students/year university-wide)**

Sunseom Scholarship GIST, Feb 2021
Awarded for outstanding creativity and research potential. **(2 students/year university-wide)**

AI Industrial Intelligence Competition Korea Artificial Intelligence Association, 2020
Excellence Award, Developed an AI voice recognition model.

TEACHING EXPERIENCE

Teaching Assistant, Linear Algebra (GS2004) GIST, Gwangju, Republic of Korea, Mar 2021 – Jun 2021
- Selected as an **undergraduate teaching assistant**, assisting in instruction and problem-solving sessions.

Coding Instructor, National Information Society Agency (NIA) Vietnam (remote), Jun 2021 – Aug 2021
- Selected as a **university representative instructor** for the ICT World Friends Korea Program
- Delivered **Python and JavaScript lectures** to international university students
- Designed and led **hands-on programming curricula**, demonstrating strong **coding proficiency**

SKILLS

Programming: Python, MATLAB, C/C++, Git, gdsfactory
Simulation: Lumerical, COMSOL, Tidy3D
Experimental: Optical instrumentation & automation; on-chip photonics measurements (nonlinear optical conversion, microcombs); free-space optical experiments; cryogenic & electrical characterization; nanofabrication (e-beam lithography, photolithography)
Languages: Korean (native), English (fluent)